# PREDICTING THE MORTALITY RATE OF NONCOMMUNICABLE DISEASES USING HEALTH, POLITICAL, AND SOCIO-ECONOMIC INDICATORS

Submitted in Partial Fulfillment of the Requirements for
Statistics 136: Introduction to Regression Analysis

Submitted by:
ANTONIO, Roger Jr. G.
CUYOS, Airam Dave L.
VARON, Carissa Elaine F.

# TABLE OF CONTENTS

ABSTRACT ..... 3

INTRODUCTION ..... 4

Objectives ..... 4

Significance of the Study ..... 4

Scope and Limitations ..... 5

REVIEW OF RELATED LITERATURE ..... 6

Regarding Noncommunicable Disease Mortality Rate: ..... 6

Regarding Agriculture: ..... 6

Regarding Air Pollution: ..... 6

Regarding Alcohol Consumption: ..... 6

Regarding Primary Care on NCDs: ..... 7

Regarding Sweetened Beverages: ..... 7

Regarding Fruit and Vegetable Consumption: ..... 7

Regarding Income Level: ..... 7

Regarding Health Expenditure: ..... 7

Regarding Obesity: ..... 8

Regarding Drinking Water Services: ..... 8

Regarding Physical Activity: ..... 8

Regarding Cholesterol: ..... 8

Regarding Tobacco: ..... 8

METHODOLOGY ..... 9

Data Collection ..... 9

Data Description ..... 9

Overview of the Modelling Procedure ..... 11

RESULTS AND DISCUSSION ..... 12

Initial Full Model ..... 12

Variable Selection ..... 12
Diagnostic Checking ..... 13
Non-normality ..... 13
Heteroskedasticity ..... 14
Multicollinearity ..... 15
Outliers and Influential Observations ..... 16
Final Model ..... 21
CONCLUSION ..... 22
REFERENCES ..... 23
APPENDIX ..... 26

**ABSTRACT**

The purpose of this research is to create a model that explains the variability of the NCD mortality rate using health, political, and socio-economic indicators. The data for this study is collected from each country based on the completeness of data regarding the indicators. The indicators include agricultural land area, air pollution, alcohol consumption, existence of NCD guidelines through primary care, proportion using at least basic drinking-water services, prevalence of insufficient physical activity, mean total cholesterol for male and female, tobacco consumption, vegetable consumption, existence of tax on sugar-sweetened beverages, fruit consumption, GNI per capita classification, health expenditure, and obesity prevalence.

The final regression model shows that the proportion using at least basic drinking-water services, prevalence of insufficient physical activity among adults aged 18+ years, fruit consumption, health expenditure, obesity prevalence, and GNI per capita classification can explain the variability of the NCD mortality rate among the countries.

The researchers suggest that policies that give more access to water, food, and healthcare should be implemented to reduce the NCD mortality rate.

# INTRODUCTION

Noncommunicable diseases (NCDs) or also known as chronic diseases are diseases that are not passed from person to person and are of long duration and slow progression. Main types of NCDs include cardiovascular diseases, cancers, respiratory diseases, diabetes and mental disorders. NCDs collectively account for the deaths of 41 million people annually, equivalent to 71% of all deaths globally (WHO, 2021). In the Philippines, for the year 2018, more than 200,000 account for deaths due to heart diseases and cancer while less than 99,000 were caused by communicable diseases (DOH, 2018).

NCDs are commonly identified as a result of a combination of genetic, physiological, environmental and behavioral factors. On an economic standpoint, non- high income countries are more susceptible to NCDs. A significant 82% of premature deaths, which are deaths of age less than 70, came from low and middle-income countries (WHO, 2021).

Although highly lethal, most NCDs are preventable. In most countries, governments have proposed and implemented restrictions on the consumption of unhealthy products such as alcohol and tobacco (CDC, 2013). In fact, the World Health Organization (WHO, 2021) has identified four major risk factors that primarily drive the cases of NCDs namely: tobacco use, physical inactivity, the harmful use of alcohol and unhealthy diets. In order to further understand what contributes to the mortality of NCDs, a wider scope of factors must be considered.

## Objectives

This study has the following objectives:

(1) Explain the variability of the mortality rate of NCDs among countries using health, political, and socio-economic indicators through a linear model.
(2) Identify significant factors that can explain the variability of the mortality rate of NCDs.
(3) Suggest solutions, recommendations, and/or strategies as to how the mortality rate of NCDs can be controlled or even lowered

## Significance of the Study

The findings of the study will be helpful for government entities by identifying factors which they can use to focus their efforts in order to combat NCDs. The study will also be beneficial to future researchers as it provides further understanding on the topic.

**Scope and Limitations**

This study is limited to studying the possible significance of these factors on the mortality rate of NCDs: Agricultural Land Area, Air Pollution, Alcohol Consumption, Existence of NCD Guidelines through Primary Care, Proportion Using at Least Basic Drinking-Water Services, Prevalence of Insufficient Physical Activity, Mean Total Cholesterol for Male and Female, Prevalence of Tobacco Use, Fruit and Vegetable Consumption, Existence of Tax on Sugar-Sweetened Beverages, GNI Per Capita Classification, Health Expenditure, and Obesity Prevalence.

Due to data limitations which include data availability and completeness of data, not all countries were included in the study. Also, only years 2016 and 2017 were utilized since these are the latest years of which data are available for all the countries.

## REVIEW OF RELATED LITERATURE

**Regarding Noncommunicable Disease Mortality Rate:**

According to the World Health Organization (2021), NCDs, which are also known as chronic diseases, are the result of a combination of genetic, physiological, environmental, and behavioural factors and may tend to be of long duration. Some types of NCDs are cardiovascular diseases, cancers, chronic respiratory diseases, and diabetes. They also found that NCDs kill 41 million people each year, equivalent to 71% of all deaths globally.

**Regarding Agriculture:**

In 2018, the World Health Organization pointed out that the food and agriculture sector is inevitably linked with NCDs. Providing broader food, agriculture and socio-economic policies will have a major impact on NCDs. It has also been stated that Agenda 2030, 2014 Rome Declaration of ICN2, and the proclamation of the Decade of Action on Nutrition acknowledge the need for action across the food system to lessen NCDs. They have also mentioned that the ministries of agriculture must regularly consider both caloric quality and quantity, making sure that policies designed to address undernutrition do not inadvertently result in overweight, obesity, high blood pressure, and NCDs.

**Regarding Air Pollution:**

According to the World Health Organization (2019), the second leading cause of death from NCDs after tobacco-smoking is air pollution. Such NCDs associated with air pollution are ischaemic heart disease, stroke, chronic obstructive pulmonary disease and lung cancer. The WHO European Region also found in 2016 that more than 550,000 deaths were attributed to the joint effects of households and ambient air pollution. In addition, in 2018, the third United Nations high-level meeting on NCDs identified household and outdoor air pollution as a risk factor for NCDs, alongside unhealthy diets, tobacco-smoking, harmful use of alcohol and physical inactivity.

**Regarding Alcohol Consumption:**

A study by Parry, Patra, and Rehm (2012) found that alcohol is linked to eight different cancers, with the risk increasing with the volume of alcohol consumed. Furthermore, alcohol use is negatively related to many cardiovascular outcomes, including hypertension, haemorrhagic stroke and atrial fibrillation. They also stated that alcohol is linked to various forms of liver disease (specifically with fatty liver, alcoholic hepatitis, and cirrhosis) and pancreatitis. Moreover, 3.4% of the global NCD-related burden of deaths, 5.0% of the net years of life lost (YLL), and 2.4% of the net disability adjusted life years (DALYs)

can be all attributed to alcohol consumption, which was also discovered being notably high for cancer and liver cirrhosis.

**Regarding Primary Care on NCDs:**

According to a study by Varghese (2019), evidence from New Zealand, Mexico, England, Wales, and countries of the Organization for Economic Cooperation and Development countries shows that reduction in cause specific premature mortality from asthma and bronchitis or emphysema, cancer, and cardiovascular disease is attributed to treatment and improvements in primary care. In completely developed health systems, most of the care for noncommunicable diseases is handled by primary care. Conversely, in poorer settings, most primary care facilities are not equipped to identify and treat NCDs.

**Regarding Sweetened Beverages:**

Diabetes, a type of non-communicable disease, is highly associated with the amount of intake of sweet food and drinks which typically contain high amounts of sugar. A study by Wang, Fung, Hang, & Yu (2014) published in the Journal of Diabetes Investigation concludes that sugar-sweetened beverages were associated with an increased risk in type 2 diabetes which is a noncommunicable disease.

**Regarding Fruit and Vegetable Consumption:**

From a study by Kjøllesdal, et.al (2016), a higher amount of intake of fruits and vegetables results in significantly lower cholesterol for men but the other way around for women but for both genders, it was then attributed to lower risks for hypertriglyceridemia, a type of noncommunicable disease.

**Regarding Income Level:**

According to the World Health Organization (2021), 15 million premature deaths due to NCDs happen per year. This refers to the deaths that occur between ages 30-69. Of these premature deaths, 85% happen in low and middle-income countries. Also, 77% of the total deaths that occur due to NCD takes place in low and middle-income countries.

**Regarding Health Expenditure:**

A study conducted by Meghani, et.al (2021) in Uganda suggests more adequate funding for Uganda to curb the rise of noncommunicable diseases. Results show that the funding constraints, like limited experts in the Health ministry, that occur in Uganda played a big role in slowing down the NCD prevention and control efforts in Uganda. With this, the risks for NCDs tend to increase in Uganda.

**Regarding Obesity:**

The World Health Organization has listed obesity as one of the modifiable risk factors of NCDs. In an article by the Ministry of Health of Saudi Arabia (2020), obesity poses risks to several noncommunicable diseases like diabetes, cardiovascular and kidney diseases, and cancer.

**Regarding Drinking Water Services:**

Water is essential to support all life. Even though basic clean drinking water is a human right, not everyone is fortunate enough to receive it. According to WHO (2019), only 90% of the global population used at least a basic service. 785 million people lack an improved drinking water source and 144 million people are dependent only on surface water. Insufficient water services not only affect the health of the body but also the ability of health services to amend diseases. Globally, 15% of patients develop infections during a hospital stay, with the proportion much greater in low-income countries with insufficient basic water services.

**Regarding Physical Activity:**

A 2013 paper studying the effect of inactivity to non-communicable diseases found that physical inactivity has a major health impact on the world. They also found that inactivity was responsible for 9% of premature mortality in 2008 which accounts for 57 million deaths. They added that if inactivity was decreased by 10%, 533,000 deaths may be averted each year (Lee, Shiroma, Lobelo et al., 2012).

**Regarding Cholesterol:**

The United States Centers for Disease Control and Prevention flagged High Cholesterol levels as a major risk factor for diabetes which is one of the major types of NCDs. Cholesterol could be categorized into High Density Lipoprotein(HDL) and non-HDL. HDL cholesterol helps remove other cholesterol from the arteries and is known to be good while non-HDL cholesterol includes those that cause blockages. This is why in this study, the researchers will focus and solely look into non-HDL cholesterol levels (US NLM, 2020).

**Regarding Tobacco:**

Tobacco usage is a common and highly dangerous risk factor to many NCDs. Globally, 14% of all NCDs deaths among adults aged 30 years and over are attributable to tobacco usage (NCD Alliance, 2019). WHO (n.d.) considers tobacco usage as an epidemic as it kills nearly 6 million people a year. Of which, 5 million are directly related to first hand usage while greater than 600,000 are the result of non-smokers being exposed to secondhand smoke.

## METHODOLOGY

### Data Collection

Datasets for all the variables are gathered from the World Health Organization, World Bank Data Catalog, and the Non-Communicable Diseases Risk Factor Collaboration which are all credible networks and organizations when it comes to providing data for the variables that the researchers would use. The dataset contains data across different years but the researchers chose to get the data from either 2016 or 2017 because all of the variables have available data from either of these two years. This also means that not all data came from the same year. Some variables have data gathered from 2016 while other variables have data gathered from 2017.

### Data Description

The table below shows all the variables that are included in the model with their respective labels, years, descriptions, and sources.

| Variable (Label) | Year | Description | Source |
|---|---|---|---|
| Non-Communicable Diseases Mortality Rate per 100,000 population (NCDmortality) | 2017 | Refers to the number of deaths caused by Non-Communicable Diseases per 100,000 population | World Health Organization |
| Agricultural Land Percentage (AgriLand) | 2017 | The percentage of the total land area that are agricultural land | World Bank |
| PM2.5 air pollution, mean annual exposure (AirPollution) | 2017 | The average annual exposure to air particles that measure at most 2.5 microns in micrograms per cubic meter | World Bank |
| Alcohol Consumption (Alcohol) | 2017 | The recorded alcohol intake per capita of ages 15 and above of all types of alcohol measured in liters of pure alcohol | World Health Organization |
| Existence of NCD guidelines through primary care (Guidelines) | 2017 | If the country has approved evidence-based national guidelines/protocols/standards for the management of NCDs. Baseline: No existence | World Health Organization |
| Proportion using at least basic drinking-water services (Water) | 2017 | The percentage of the population that gathers and drinks from a source that can be travelled within 30 minutes of round trip | World Health Organization |
| Prevalence of insufficient physical | 2016 | The percentage of the population who have at most 150 minutes of moderate | World Health Organization |

| | | | |
|---|---|---|---|
| activity among adults aged 18+ years (Physical) | | physical activity per week, or at most 75 minutes of intense physical activity per week, or equivalent | |
| Mean total cholesterol [Non-HDL] for Male (CholesterolM) | 2017 | The average bad cholesterol intake in millimoles per liter for males | NCD Risk Factor Collaboration |
| Mean total cholesterol [Non-HDL] for Female (CholesterolF) | 2017 | The average bad cholesterol intake in millimoles per liter for females | NCD Risk Factor Collaboration |
| Prevalence of current tobacco use (Tobacco) | 2016 | The percentage of adults that currently use tobacco products | World Bank |
| Vegetable Consumption (Vegetable) | 2017 | The average vegetable consumption per capita per year measure in kilograms | UN Food and Agricultural Organization |
| Existence of tax on sugar-sweetened beverages (Tax) | 2017 | If the country has implemented taxes on sugar-sweetened beverages. Baseline: No existence | World Health Organization |
| Fruit Consumption (Fruit) | 2017 | The average fruit consumption per capita per year measure in kilograms | UN Food and Agricultural Organization |
| GNI Per Capita Classification (GNIClass) | 2017 | Classification of the Gross National Income per capita based on the World Bank. Low Income- $1005 or less; Lower-Middle Income- $1006-$3995; Upper-Middle Income- $3956-$12235; High Income-$12236 or more. Baseline: Low Income. | World Bank |
| Health Expenditure (Health) | 2017 | The current health expenditure per capita | World Bank |
| Obesity Prevalence (Obesity) | 2016 | The percentage of adults with at least a BMI of 30. | World Health Organization |

**Overview of the Modelling Procedure**

The procedure starts from the full model with the dependent variable (NCD Mortality) and the 15 independent variables. The full model would then undergo an overall test and individual regressors test to see the significance of the full model and the individual regressors.

Next, the researchers will move to variable selection by using All Possible Regressions and Automatic Search Procedures. In APR, the researchers would choose the best possible subset of variables based on the Adjusted R Squared and Mallows Cp values. On the other hand for ASP, the researchers would execute stepwise, forward, and backward selection processes to see the variables that will be added or left out. The reduced model would come from the comparison of the results of both the APR and ASP.

The researchers will then move to diagnostic checking to check and remedy the possible issues that might arise from the reduced model. The issues that will be addressed are Nonlinearity, Non-normality, Heteroskedasticity, and Multicollinearity. After diagnostic checking, if some regressors are added, removed, or transformed, the more reduced/modified model would undergo a final diagnostic check, an overall test, and individual regressors test to again check the significance of the final model and the remaining individual regressors.

Finally, the researchers would look for possible outliers and influential observations in the data by checking the leverage values, and plots of the studentized residuals, cook's distance, DFFITS, and DFBETAS. The researchers will then look for the reasons on why the chosen observations are the outliers and influential observations and provide some remedy if necessary to have a better model.

## RESULTS AND DISCUSSION

### Initial Full Model

The initial full model consists of the *NCDmortality* as the dependent variable and 15 regressors. OUTPUT 1 shows the output of the summary of the full model.

The output shows significant overall F-test with a very low p-value. This means that at least one of the regressors contributes significantly to the full model at alpha=0.05. The output also shows an R-squared value of 0.7114 which indicates that 71.14% of the variability in *NCDmortality* can be explained by the regressors in the model, and an adjusted R-squared value of 0.653 for the adjustment in the number of variables. By looking at the individual tests for each regressors, only *Physical, Vegetable, Health, Obesity,* and *GNIClassLowMiddle* show significant results at alpha=0.05. This cannot be concluded right away for the final model as the researchers would still have to consider variable selection procedures in the next steps.

### Variable Selection

The researchers first performed Automatic Search Procedures, specifically the Forward, Backward, and Stepwise Procedures. The researchers use a 0.05 significance level for entry and staying. The outputs of the three procedures are shown in OUTPUTS 2, 3, and 4.

Forward Selection results in a model that consists of the following regressors: *GNIClass, Obesity, Water, Tobacco, Health, Physical,* and *Fruit.*

Backward Selection results in a model that consists of the following regressors: *GNIClass, Obesity, Water, Health, Physical,* and *Fruit.*

Stepwise Selection results in a model that consists of the following regressors: *GNIClass, Obesity, Water, Health, Physical,* and *Fruit.*

Following the results of the three procedures, the researchers consider the model with regressors: *GNIClass, Obesity, Water, Health, Physical,* and *Fruit* which are both results of the backward and stepwise procedures.

The researchers then performed an All Possible Regressions approach by comparing the Adjusted R-Squared and Mallows' Cp of all the possible subset of regressors in the full model. The summary of the APR is shown in OUTPUTS 5 and 6 where the best possible subsets are compiled.

By looking at model 6, it is the same as the model that the researchers chose after the automatic search procedures and by checking its Adjusted R-Squared value, it is relatively high compared to the other models. The researchers can also consider model number 7 as it also has a close adjusted-R squared value.

Based on the outputs of Models 6 and 7 as shown in OUTPUTS 7 and 8, the researchers chose model 6 compared to model 7 because model 7 shows an insignificant regressor which is *Vegetable* unlike

in Model 6 where all the regressors are significant except for some of the dummy variables under GNIClass but the researchers retained them since one of the dummy variables is still significant. Also, both APR and ASP suggest that *GNIClass* should be included.

The new fitted regression model is:

$$\widehat{NCDmortality} = 881.3 - 3.506(Water) - 3.834(Physical) - 0.6892(Fruit) - 0.03223(Health) + 9.893(Obesity) - 112.7(GNIClassHigh) + 107.3(GNIClassLowMiddle - 13.25(GNIUpperMiddle)$$

The researchers proceed to Diagnostic Checking.

**Diagnostic Checking**

**Non-normality**

To detect non-normality in the model, the researchers generate a normal probability plot as seen in FIGURE 1. The plot shows no significant deviation from normality except from some outliers at the right side of the plot.

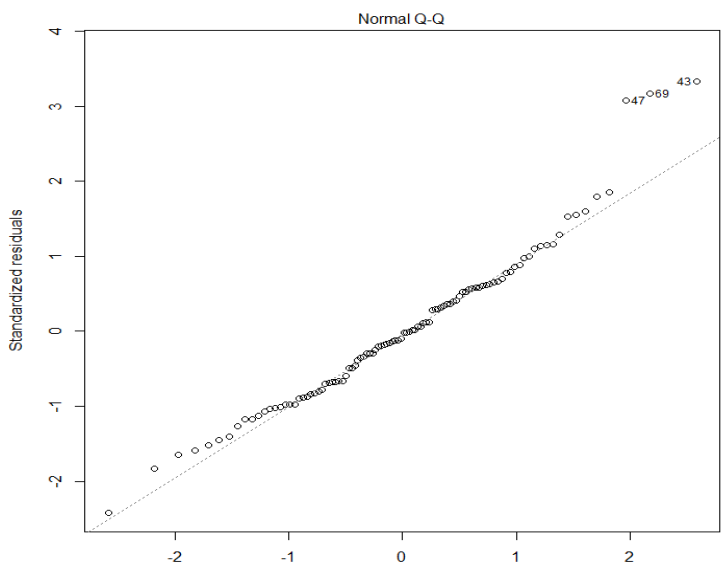


FIGURE 1: Normal Probability Plot

To formally test its normality, the researchers use Anderson-Darling Test and Shapiro-Wilk Test. In these tests for normality, a non-significant result is desired. Both tests are done at $\alpha = 0.05$.Based on the output on OUTPUT 9, Anderson-Darling Test shows a non-significant result with p-value of 0.1415. However, the Shapiro-Wilk Test, the more powerful test, shows a significant result with p-value of 0.02115 which is mainly because of the outliers shown in the Normal Probability Plot. With that, the researchers decided not to address the problem in the Shapiro-Wilk Test despite showing a significant result.

**Heteroskedasticity**

Based on the individual residual plots of the regressors in FIGURE 2, non-constancy of variance is exhibited in the proportion of using at least basic drinking-water services and health expenditure.

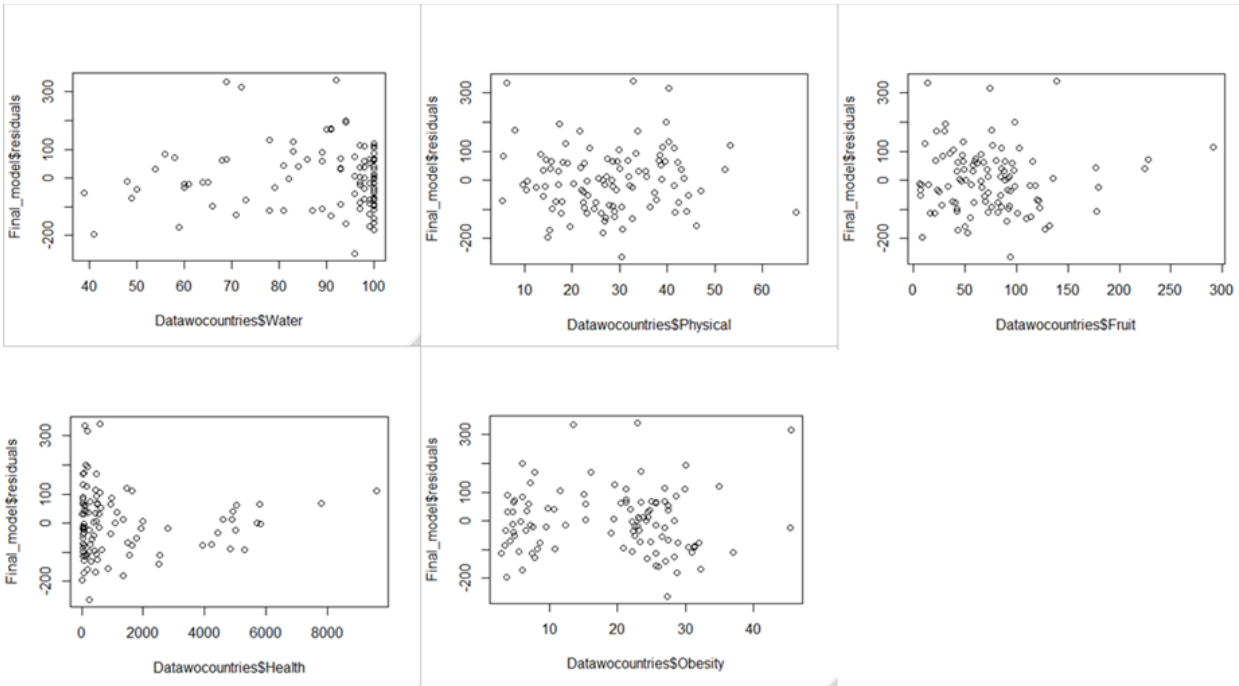


FIGURE 2: Individual Residual Plots per Regressor

To formally check the existence of heteroskedasticity, the researchers conduct the Breusch-Pagan Test. Based on the output on OUTPUT 10, the test shows a significant p-value of 0.02066. This suggests a slight issue of heteroskedasticity in the model.

Some of the factors that explain the heteroskedasticity of the model are the uneven occurrences in the observations particularly in the categorical variables of the data and the notably distinct variances of the residuals for the categorical variables. Under the GNIClass variable, 33 countries belong to the high class, 14 countries belong to the low class, 29 countries belong to the upper middle class, and 26 countries belong to the low middle class. Since some of the categories have much lower or much higher than the average of the 102 countries, this may be considered as a factor of heteroskedasticity. Looking at the variances of the residuals for the categorical variables in OUTPUT 11, the variances are relatively far from each other. The value of the variance for the residuals of the low class is at 7500.926. The value of the variance for the residuals of the low middle class is at 20146.81. The value of the variance for the residuals of the upper middle class is at 9610.752. Lastly, the value of the variance for the residuals of the high class is at 9431.728.

Despite these factors, the researchers deemed to accept the model in spite of the p-value because it is still relatively close to the level of significance $\alpha = 0.05$. In addition to this, it can also be noticed in the

summary of the final model in OUTPUT 7 that the standard errors of the variables are relatively low. Hence, the researchers choose to not address the problem of heteroskedasticity.

**Multicollinearity**

To detect multicollinearity, the researchers first checked the t-tests for individual parameters. As mentioned and reasoned out before, there are two non-significant variables that arise from the model which are *GNIClassHigh* and *GNIClassUpperMiddle* but the researchers chose to not address the issue due to the same reason mentioned at the variable selection part. The researchers also checked on the Variance Inflation Factors. Based on OUTPUT 12, it can be observed that there is no VIF that exceeds 10 which means that there are no signs of serious multicollinearity. However, the VIF for *GNIClass* is notable with a VIF of 9.158297.

**Outliers and Influential Observations**

To detect Outliers and Influential Observations, the researchers will use multiple methods. Specifically, the leverage, studentized Residuals, Cook's Distance, DFFITS, and DFBETAS will be used to flag significant observations.

Looking into the leverages of each country, 10 countries were found to be leverage points as s, namely the following:

Dominican Republic (24), Kiribati (43), Kuwait (44), Mali (54), Nepal (63), Norway (68), Rwanda (76), Samoa (77), Sao Tome and Principe (78), Switzerland (89)

Looking into the studentized residuals of the observations in FIGURE 3, there is no country that went over the threshold but a country worth looking into is Tunisia (93) which is the only country that has a studentized residual greater than abs(2).

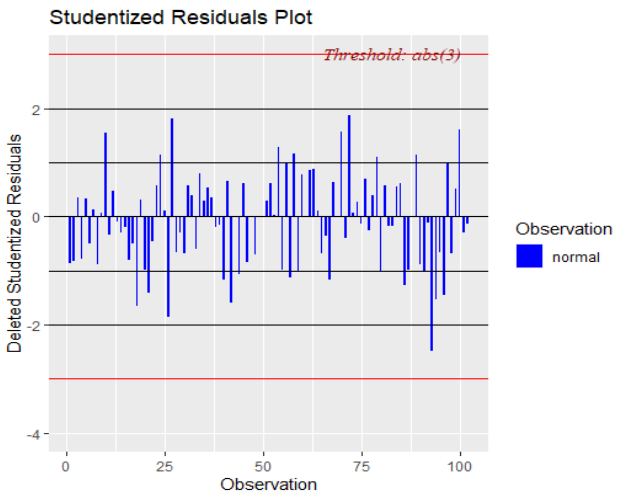


FIGURE 3: Studentized Residual Plots

Combining the result of the leverage and studentized Residuals of the countries, although the 10 aforementioned countries were found to be leverage points, they are not found to be outliers by their studentized residuals.

Looking into the Cook's Distance (FIGURE 4) and DFFITS (FIGURE 5) values for the observations, The results are fairly similar. Countries that were higher than the threshold were: Dominican Republic (24) , Ethiopia (26), Kiribati (43), Lesotho (47), Oman (69), Philippines (72), Switzerland (89), Tunisia (93). These countries are flagged as possibly influential. It is also notable that among all observations, Kiribati (43) was found to be the most influential through looking at the charts.

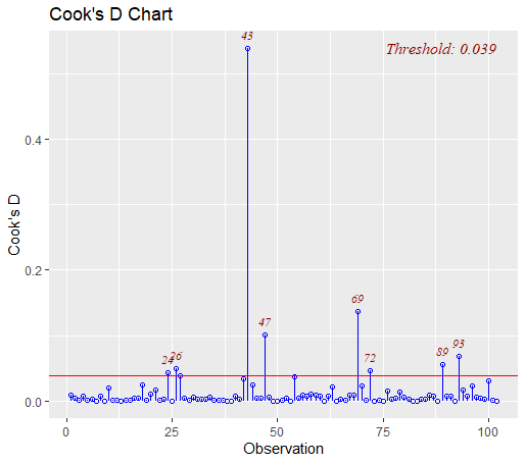


FIGURE 4: Cook's D Chart

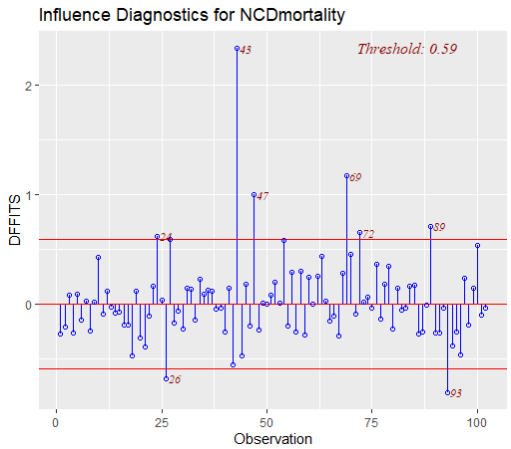


FIGURE 5: DFFITS Plot

Looking into DFBETAS in FIGURES 6, 7, and 8 to find influential observations:

Countries that have significant influence and their explanation are as follows:

Dominican Republic (24) has high influence on Fruit(+)

- Dominican Republic has the highest fruit consumption among all observations (291.01)
- Dominican Republic is a country in the Carribean that is known for fruit production. In 2017, they were ranked fourth in the importance of fruits worldwide (IICA, 2017).

Kiribati (43) has high influence on Physical(+), Water(-), Obesity(+)

- Kiribati has high physical inactivity percentage with 40.4 that is greater than the third quartile of the data [Q3: 36.23, max: 67.00]
- Kiribati has low water proportion of water services with 72 that is less than the first quartile of the data [min: 39.00,Q1: 81.25]
- Kiribati has the highest obesity prevalence among all observations (45.60)
- Kiribati is a country in Oceania where obesity is truly prevalent and it is partially due to lack of physical activity. Being an archipelago, there is a geographical disparity where outer islands are particularly vulnerable to lack of resources and services (Eme, Burlingame, Kim et al., 2020).

Lesotho (47) has high influence on Physical(-),

- Lesotho has low physical inactivity percentage with 6.3 that is less than the first quartile of the data [min: 5.50,Q1: 18.73]
- Lesotho is a low-income mountainous country in South Africa that has a high percentage of its population engaging in agriculture (FAO). It also had a very low count of vehicles per 1,000 population in 2014 being only 4 (NationMaster, n.d.). This may be a reason why physical inactivity is not prevalent in the country.

Oman (69) has high influence on Fruit(+), Health(-)

- Oman has high fruit consumption with 138.93 that is greater than the third quartile of the data [Q3: 93.81, max: 1, 291.01]
- Oman has around average health expenditure with 601.22 [Median: 393.17, Mean: 1246.22] but what makes Oman have high influence is that among countries with the same GNIClass (High), Oman has the lowest health expenditure. (High GNI health mean: 3361.9)
- The fruit and vegetable consumption is very high in Oman. Its fruit and vegetable production even nearly tripled from 2016-2017 (MordorIntelligence, n.d.).

Switzerland (89) has high influence on Health(+)

- Switzerland has the highest health expenditure among all observations (9606.35)
- Switzerland is known to have the best healthcare system in the world (SwissLife, 2011). It is only reasonable for it to have higher expenditure.

Going back to the previous charts where Kiribati was found to be the most influential, it is explained by it having multiple significant factors explained above. As all of the influential observations that were discussed are legitimate, no observation will be omitted from the dataset.

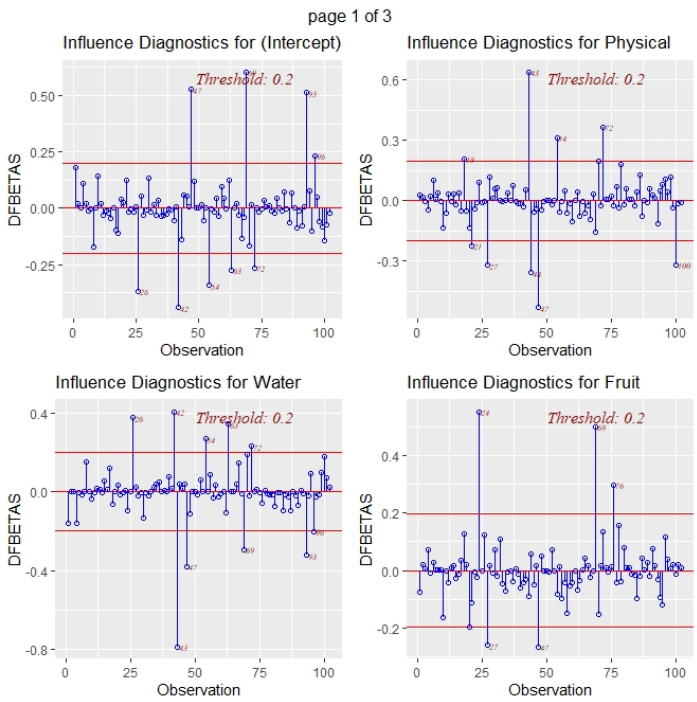


FIGURE 6: DFBETAS Plot Page 1

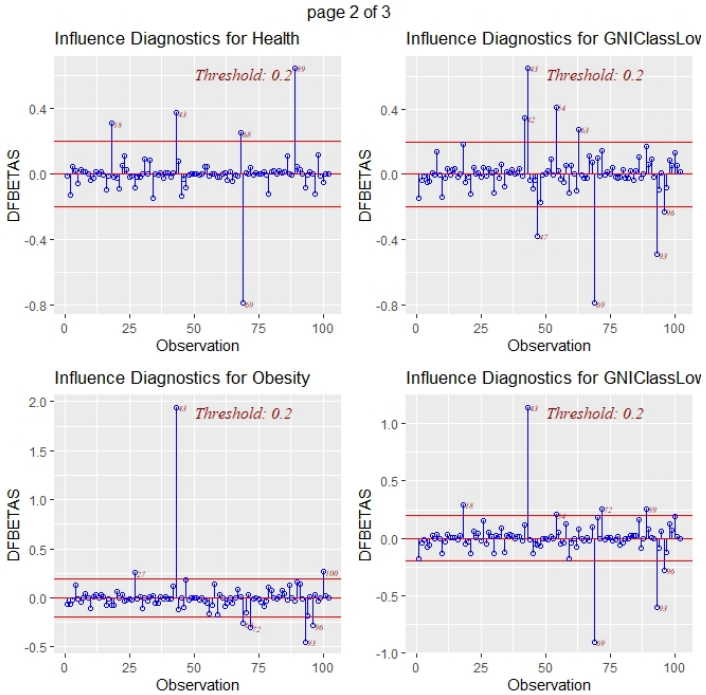


FIGURE 7: DFBETAS Plot Page 2

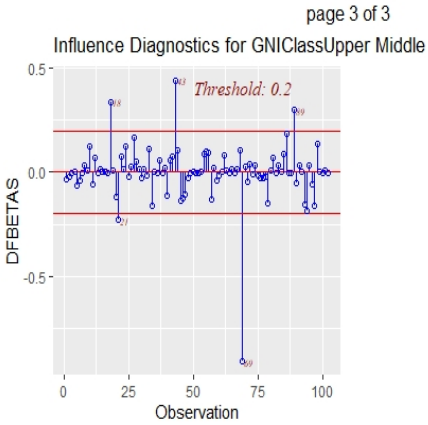


FIGURE 8: DFBETAS Plot Page 3

**Final Model**

Based on what transpired in diagnostic checking, the final model is still the same from what was obtained after variable selection since no regressor was added, removed, or transformed during the process of diagnostic checking. However, there are slight issues of heteroskedasticity and non-normality present in the model which was explained earlier. Thus, the final model is still given by:

$$\widehat{NCDmortality} = 881.3 - 3.506(Water) - 3.834(Physical) - 0.6892(Fruit) - 0.03223(Health) + 9.893(Obesity) - 112.7(GNIClassHigh) + 107.3(GNIClassLowMiddle) - 13.25(GNIUpperMiddle)$$

Interpretation of the model:

- A unit increase in *Water* decreases *NCDmortality* by 3.506 on average keeping all the other variables constant.
- A unit increase in *Physical* decreases *NCDmortality* by 3.834 on average keeping all the other variables constant.
- A unit increase in *Fruit* decreases *NCDmortality* by 0.6892 on average keeping all the other variables constant.
- A unit increase in *Health* decreases *NCDmortality* by 0.03223 on average keeping all the other variables constant.
- A unit increase in *Obesity* increases *NCDmortality* by 9.893 on average keeping all the other variables constant.
- A unit increase in *GNIClassHigh* increases *NCDmortality* by 112.7 less than *GNIClassLow* on average keeping all the other variables constant.
- A unit increase in *GNIClassLowMIddle* increases *NCDmortality* by 107.3 than *GNIClassLow* on average keeping all the other variables constant.
- A unit increase in *GNIUpperMiddle* increases *NCDmortality* by 13.25 less than *GNIClassLow* on average keeping all the other variables constant.

## CONCLUSION

The researchers have found a model explaining 67.48% of the variability in the NCD mortality rate can be generated with significant regressors. These significant regressors are namely the proportion using at least basic drinking-water services, prevalence of insufficient physical activity among adults aged 18+ years, fruit consumption, health expenditure, obesity prevalence, and GNI per capita classification.

Based on the significant regressors in the model, the researchers encourage governments to push forward policies that give more access to water, food, and healthcare to reduce the NCD mortality rate.

For the future studies about this topic, the researchers recommend to include data on other specific variables such as data regarding depression, COVID-19 cases, response against COVID-19, and other factors regarding health.

## REFERENCES


Centers for Disease Control and Prevention-Department of Health and Human Services. (2013). Overview of Noncommunicable Diseases and Related Risk Factors.

Cghe. (n.d.). Global Alliance for Chronic Diseases. GACD. https://www.gacd.org/research/what-are-ncds/chronic-diseases-fact-sheet.

Department of Health-Epidemiology Bureau Department of Health . (2018). The 2018 Philippine Health Statistics. Philippines.

Eme, P. E., Burlingame, B., Kim, N. D., Foliaki, S., Wham, C., & Douwes, J. (2020, July 11). Obesity measures in the Kiribati population: a need to reclassify body mass index cut-points. BMC Public Health. https://bmcpublichealth.biomedcentral.com/articles/10.1186/s12889-020-09217-z.

Kjøllesdal, M. et al. (2016). Consumption of fruits and vegetables and associations with risk factors for non-communicable diseases in the Yangon region of Myanmar: a cross-sectional study. *BMJ Open, 6(8).* Retrieved from https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5013453/

Lee, I.-M., Shiroma, E. J., Lobelo, F., Puska, P., Blair, S. N., Katzmarzyk, P. T., & Lancet Physical Activity Series Working Group. (2012, July 21). Effect of physical inactivity on major non-communicable diseases worldwide: an analysis of burden of disease and life expectancy. Lancet (London, England). https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3645500/.

Meghani, A. et al. (2021). Curbing the Rise of Noncommunicable Diseases in Uganda: Perspectives of Policy Actors. *Global Health: Science and Practice March 2021, 9(1)*. Retrieved from https://www.ghspjournal.org/content/9/1/149

NationMaster. (n.d.). Lesotho Labor Stats. Country Info. https://www.nationmaster.com/country-info/profiles/Lesotho/Labor.

Oman Fruits and Vegetables Market - Growth, Trends, COVID-19 Impact, and Forecasts (2021 - 2026). (n.d.). https://www.mordorintelligence.com/industry-reports/fruits-and-vegetables-industry-industry.

Parry, C. D., Patra, J., & Rehm, J. (2012, October). *Alcohol consumption and non-communicable diseases: epidemiology and policy implications*. Addiction (Abingdon, England). https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3174337/

Saudi Arabian Ministry of Health (2020). Non-Communicable Diseases. *Educational Content.* Retrieved from https://www.moh.gov.sa/en/HealthAwareness/EducationalContent/Diseases/NonCommunicable/Pages/Obesity.aspx

The Avocado Crop: Great Potential for the Dominican Republic. IICA.INT. (2017). https://www.iica.int/en/node/17304#:~:text=There%20are%20currently%2021%2C875%20hectares,the%20avocado%20varieties%20are%20local.

The Dominican Republic is the second biggest producer of avocados in the world. FreshPlaza. (2018, April 4). https://www.freshplaza.com/article/2192196/the-dominican-republic-is-the-second-biggest-producer-of-avocados-in-the-world/.

The Swiss Healthcare System: a Model for Others? Swiss Life Group. (2011, June 3). https://www.swisslife.com/en/home/media/media-releases/news-archiv/the_swiss_healthcare.html

Tobacco Use. NCD Alliance. (2019, October 17). https://ncdalliance.org/why-ncds/ncd-prevention/tobacco-use.

U.S. National Library of Medicine. (2020, October 2). Cholesterol Levels: What You Need to Know. MedlinePlus. https://medlineplus.gov/cholesterollevelswhatyouneedtoknow.html#:~:text=Total%20cholesterol%20%2D%20a%20measure%20of,density%20lipoprotein%20(HDL)%20cholesterol.&text=Non%2DHDL%20%2D%20this%20number%20is,total%20cholesterol%20minus%20your%20HDL.

UNICEF Pacific. (2017, December). Situation Analysis of Children in Kiribati.

Varghese, C., Nongkynrih, B., Onakpoya, I., McCall, M., Barkley, S., & Collins, T. E. (2019, January 28).

*Better health and wellbeing for billion more people: integrating non-communicable diseases in primary care*. BMJ (Clinical research ed.). https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6349006/

Wang, M., Fung, M., Hang, L., & Yu, R. (2014). Association between sugar-sweetened beverages and type 2 diabetes: A meta-analysis. *Journal Of Diabetes Investigation, 6(3).* Retrieved from https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4420570/

World Health Organization. (2019). Drinking-water. World Health Organization. https://www.who.int/news-room/fact-sheets/detail/drinking-water#:~:text=Key%20facts,at%20least%20a%20basic%20service.

World Health Organization. (n.d.). Eastern Mediterranean Region Tobacco on Noncommunicable Diseases. World Health Organization. http://www.emro.who.int/noncommunicable-diseases/causes/tobacco-use.html.

World Health Organization (2021). Noncommunicable diseases. Retrieved from https://www.who.int/news-room/fact-sheets/detail/noncommunicable-diseases

World Health Organization. (2018). *Noncommunicable diseases: what ministries of agriculture need to know*. World Health Organization. https://apps.who.int/iris/handle/10665/272639

World Health Organization. (2019, April 8). *Noncommunicable diseases and air pollution (2019)*. World Health Organization. https://www.euro.who.int/en/health-topics/environment-and-health/air-quality/publications/2019/noncommunicable-diseases-and-air-pollution-2019.

## APPENDIX

```
Overall F-test
Coefficients:
                        Estimate   Std. Error  t value Pr(>|t|)
(Intercept)             851.731523 165.156676   5.157 1.64e-06 ***
Water                    -2.705914   1.573555  -1.720  0.08918 .
Physical                 -4.094429   1.330966  -3.076  0.00283 **
CholesterolM            -55.244665 124.478494  -0.444  0.65832
CholesterolW             47.694452 133.478645   0.357  0.72175
Tobacco                   1.938577   1.415197   1.370  0.17439
Vegetable                -0.424831   0.202837  -2.094  0.03923 *
TaxYes                  -39.284090  25.557227  -1.537  0.12803
Fruit                    -0.511102   0.273767  -1.867  0.06540 .
Health                   -0.032339   0.009941  -3.253  0.00165 **
Obesity                   9.055974   1.814680   4.990 3.21e-06 ***
GNIClassHigh            -57.528804  85.722821  -0.671  0.50400
GNIClassLow Middle      104.095164  49.196805   2.116  0.03731 *
GNIClassUpper Middle     15.673365  69.088694   0.227  0.82108
AgriLand                  0.093980   0.645593   0.146  0.88461
AirPollution              0.379635   0.776539   0.489  0.62620
Alcohol                  -3.620019   4.938068  -0.733  0.46555
GuidelinesYes           -26.687490  24.717860  -1.080  0.28337
---
   Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 112.8 on 84 degrees of freedom
Multiple R-squared:  0.7114,   Adjusted R-squared:  0.653
F-statistic: 12.18 on 17 and 84 DF,  p-value: 3.917e-16
```

OUTPUT 1: Summary of the Initial Full Model

```
Selection Summary
--------------------------------------------------------------------------
  Variable                  Adj.
Step  Entered     R-Square    R-Square     C(p)         AIC         RMSE
--------------------------------------------------------------------------
1    GNIClass      0.4440      0.4270    63.8286    1310.5236    144.9197
2    Obesity       0.5357      0.5166    39.1300    1294.1330    133.1076
3    Water         0.5845      0.5628    26.9467    1284.8224    126.5827
4    Tobacco       0.6177      0.5936    19.2640    1278.3112    122.0474
5    Health        0.6427      0.6161    13.9804    1273.4054    118.6109
6    Physical      0.6677      0.6392     8.7070    1268.0088    115.0007
7    Fruit         0.6829      0.6519     6.2921    1265.2426    112.9539
--------------------------------------------------------------------------
Predictors: "GNIClass" "Obesity"  "Water"    "Tobacco"  "Health"   "Physical" "Fruit"
```

OUTPUT 2: Forward Selection Output

Elimination Summary

| Step | Variable Removed | R-Square | Adj. R-Square | C(p) | AIC | RMSE |
|---|---|---|---|---|---|---|
| 1 | AgriLand | 0.7113 | 0.657 | 12.0212 | 1269.6658 | 112.1238 |
| 2 | CholesterolW | 0.7109 | 0.6605 | 10.1413 | 1267.8115 | 111.5496 |
| 3 | CholesterolM | 0.7105 | 0.6639 | 8.2659 | 1265.9625 | 110.9888 |
| 4 | AirPollution | 0.7097 | 0.6668 | 6.5028 | 1264.2489 | 110.5114 |
| 5 | Alcohol | 0.7058 | 0.6661 | 5.6394 | 1263.6116 | 110.6254 |
| 6 | Guidelines | 0.7009 | 0.6644 | 5.0442 | 1263.2712 | 110.9077 |
| 7 | Tobacco | 0.6965 | 0.6631 | 4.3341 | 1262.7716 | 111.1108 |
| 8 | Tax | 0.6853 | 0.6545 | 5.5945 | 1264.4687 | 112.5262 |
| 9 | Vegetable | 0.6748 | 0.6468 | 6.6444 | 1265.8097 | 113.7677 |

Predictors: "GNIClass" "Obesity" "Water" "Health" "Physical" "Fruit"

OUTPUT 3: Backward Selection Output

Stepwise Selection Summary

| Step | Added/ Variable | Removed | R-Square | Adj. R-Square | C(p) | AIC | RMSE |
|---|---|---|---|---|---|---|---|
| 1 | GNIClass | addition | 0.444 | 0.427 | 63.8290 | 1310.5236 | 144.9197 |
| 2 | Obesity | addition | 0.536 | 0.517 | 39.1300 | 1294.1330 | 133.1076 |
| 3 | Water | addition | 0.584 | 0.563 | 26.9470 | 1284.8224 | 126.5827 |
| 4 | Tobacco | addition | 0.618 | 0.594 | 19.2640 | 1278.3112 | 122.0474 |
| 5 | Health | addition | 0.643 | 0.616 | 13.9800 | 1273.4054 | 118.6109 |
| 6 | Physical | addition | 0.668 | 0.639 | 8.7070 | 1268.0088 | 115.0007 |
| 7 | Fruit | addition | 0.683 | 0.652 | 6.2920 | 1265.2426 | 112.9539 |
| 8 | Tobacco | removal | 0.675 | 0.647 | 6.6440 | 1265.8097 | 113.7677 |

Predictors: "GNIClass" "Obesity" "Water" "Health" "Physical" "Fruit"

OUTPUT 4: Stepwise Selection Output

Best Subsets Regression

| Model Index | Predictors |
|---|---|
| 1 | GNIClass |
| 2 | Obesity GNIClass |
| 3 | Water Obesity GNIClass |
| 4 | Water Tobacco Obesity GNIClass |
| 5 | Physical Fruit Health Obesity GNIClass |
| 6 | Water Physical Fruit Health Obesity GNIClass |
| 7 | Water Physical Vegetable Fruit Health Obesity GNIClass |
| 8 | Water Physical Vegetable Tax Fruit Health Obesity GNIClass |
| 9 | Water Physical Tobacco Vegetable Tax Fruit Health Obesity GNIClass |
| 10 | Water Physical Tobacco Vegetable Tax Fruit Health Obesity GNIClass Guidelines |
| 11 | Water Physical Tobacco Vegetable Tax Fruit Health Obesity GNIClass Alcohol Guidelines |
| 12 | Water Physical Tobacco Vegetable Tax Fruit Health Obesity GNIClass AirPollution Alcohol Guidelines |
| 13 | Water Physical CholesterolM Tobacco Vegetable Tax Fruit Health Obesity GNIClass AirPollution Alcohol Guidelines |
| 14 | Water Physical CholesterolM CholesterolW Tobacco Vegetable Tax Fruit Health Obesity GNIClass AirPollution Alcohol Guidelines |
| 15 | Water Physical CholesterolM CholesterolW Tobacco Vegetable Tax Fruit Health Obesity GNIClass AgriLand AirPollution Alcohol Guidelines |

OUTPUT 5: APR Output Part 1

```
  Subsets Regression Summary
-----------------------------------------------------
                      Adj.        Pred
Model    R-Square    R-Square    R-Square     C(p)
-----------------------------------------------------
1        0.4440      0.4270      0.4019     63.8286
2        0.5357      0.5166      0.4712     39.1300
3        0.5845      0.5628      0.5184     26.9467
4        0.6177      0.5936      0.5538     19.2640
5        0.6503      0.6243      0.5579     11.7670
6        0.6748      0.6468      0.5804      6.6444
7        0.6853      0.6545      0.5829      5.5945
8        0.6965      0.6631      0.5902      4.3341
9        0.7009      0.6644       0.581      5.0442
10        0.7058      0.6661      0.5803      5.6394
11        0.7097      0.6668      0.5706      6.5028
12        0.7105      0.6639      0.5627      8.2659
13        0.7109      0.6605       0.549     10.1413
14        0.7113      0.6570      0.5401     12.0212
15        0.7114      0.6530      0.5285     14.0000
-----------------------------------------------------
```

OUTPUT 6: APR Output Part 2

```
Residuals:
  Min      1Q  Median      3Q     Max
-262.15  -75.68   -6.74   64.27  340.90

Coefficients:
                       Estimate  Std. Error  t value Pr(>|t|)
 (Intercept)           8.813e+02  8.279e+01  10.645  < 2e-16 ***
 Water                -3.506e+00  1.325e+00  -2.646 0.009577 **
 Physical             -3.834e+00  1.232e+00  -3.112 0.002469 **
 Fruit                -6.892e-01  2.480e-01  -2.779 0.006606 **
 Health               -3.223e-02  9.256e-03  -3.482 0.000761 ***
 Obesity               9.893e+00  1.675e+00   5.908 5.65e-08 ***
 GNIClassHigh         -1.127e+02  7.735e+01  -1.457 0.148362
 GNIClassLow Middle    1.073e+02  4.820e+01   2.227 0.028381 *
 GNIClassUpper Middle -1.325e+01  6.527e+01  -0.203 0.839590
---
   Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 113.8 on 93 degrees of freedom
Multiple R-squared:  0.6748,   Adjusted R-squared:  0.6468
F-statistic: 24.12 on 8 and 93 DF,  p-value: < 2.2e-16
```

OUTPUT 7: Summary of Model 6

```
Coefficients:
                      Estimate Std. Error t value Pr(>|t|)
(Intercept)           8.530e+02  8.347e+01  10.220  < 2e-16 ***
Water                -2.789e+00  1.374e+00  -2.030 0.045225 *
Physical             -4.008e+00  1.223e+00  -3.278 0.001475 **
Vegetable            -3.192e-01  1.824e-01  -1.750 0.083411 .
Fruit                -6.527e-01  2.462e-01  -2.651 0.009451 **
Health               -3.359e-02  9.188e-03  -3.656 0.000427 ***
Obesity               1.005e+01  1.659e+00   6.058 2.98e-08 ***
GNIClassHigh         -1.186e+02  7.658e+01  -1.549 0.124738
GNIClassLow Middle    1.015e+02  4.779e+01   2.125 0.036288 *
GNIClassUpper Middle -2.166e+01  6.474e+01  -0.335 0.738658
---
   Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 112.5 on 92 degrees of freedom
Multiple R-squared:  0.6853,   Adjusted R-squared:  0.6545
F-statistic: 22.26 on 9 and 92 DF,  p-value: < 2.2e-16
```

OUTPUT 8: Summary of Model 7

```
Shapiro-Wilk normality test

data:  Final_model$residuals
W = 0.97027, p-value = 0.02115


Anderson-Darling normality test

data:  Final_model$residuals
A = 0.56325, p-value = 0.1415
```

OUTPUT 9: Output of Anderson-Darling Test and Shapiro-Wilk Test

```
studentized Breusch-Pagan test

data:  Final_model
BP = 18.077, df = 8, p-value = 0.02066
```

OUTPUT 10: Output of Breusch-Pagan Test

```
var(Final_model$residuals[Datawocountries$GNIClass == "Low"])
[1] 7500.926
var(Final_model$residuals[Datawocountries$GNIClass == "Low Middle"])
[1] 20146.81
var(Final_model$residuals[Datawocountries$GNIClass == "Upper Middle"])
[1] 9610.752
var(Final_model$residuals[Datawocountries$GNIClass == "High"])
[1] 9431.728
```

OUTPUT 11: Variances of Each Category in GNIClass

```
G            VIF   Df
Water     3.568215  1
Physical  1.565280  1
Fruit     1.143119  1
Health    2.515464  1
Obesity   2.317211  1
GNIClass  9.158297  3
```

OUTPUT 12: Variance Inflation Factors of Regressors

```
lev_res<-ols_leverage(Final_model)
lev_res[lev_res>2*(7/102)]
[1] 0.2291711 0.3038939 0.1655192 0.1686922 0.1963017 0.1572638 0.2167072 0.2252965 0.1684363 0.2763346
Data[which(lev_res>2*(7/102)),1]
[1] Dominican Republic    Kiribati              Kuwait                Mali                  Nepal
[6] Norway                Rwanda                Samoa                 Sao Tome and Principe Switzerland
```

OUTPUT 13: Leverage Points and their Corresponding Leverage Values